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0580/43

October/November 2019

2 hours 30 minutes

Additional Materials: Electronic calculator Geometrical instruments
 Tracing paper (optional)

READ THESE INSTRUCTIONS FIRST

DO **NOT** WRITE IN ANY BARCODES.

For π , use either your calculator value or 3.142.

The total of the marks for this paper is 130.

This document consists of **19** printed pages and **1** blank page.

- 1 (a) In a cycling club, the number of members are in the ratio males : females = 8 : 3.
The club has 342 females.

(i) Find the total number of members.

..... [2]

(ii) Find the percentage of the total number of members that are female.

..... % [1]

- (b) The price of a bicycle is \$1020.
Club members receive a 15% discount on this price.

Find how much a club member pays for this bicycle.

\$ [2]

- (c) In 2019, the membership fee of the cycling club is \$79.50 .
This is 6% more than last year.

Find the **increase** in the cost of the membership.

\$ [3]

- (d) Asif cycles a distance of 105 km.
 On the first part of his journey he cycles 60 km in 2 hours 24 minutes.
 On the second part of his journey he cycles 45 km at 20 km/h.

Find his average speed for the whole journey.

..... km/h [4]

- (e) Bryan invested \$480 in an account 4 years ago.
 The account pays compound interest at a rate of 2.1% per year.
 Today, he uses some of the money in this account to buy a bicycle costing \$430.

Calculate how much money remains in his account.

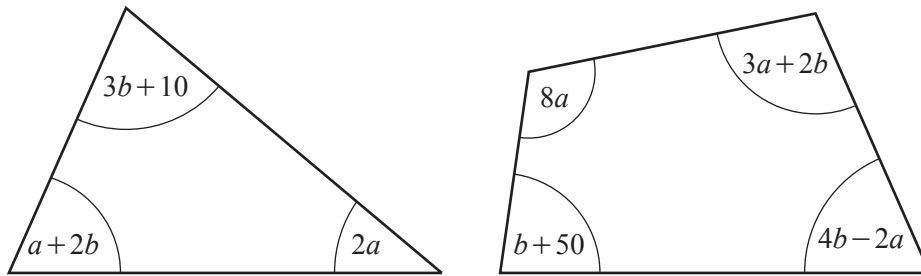
\$ [3]

- (f) The formula $s = \frac{1}{2}at^2$ is used to calculate the distance, s , travelled by a bicycle.

When $a = 3$ and $t = 10$, each correct to the nearest integer, calculate the lower bound of the distance, s .

..... [2]

- 2 (a) The diagram shows a triangle and a quadrilateral.
All angles are in degrees.



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- (i) For the triangle, show that $3a + 5b = 170$.

[1]

- (ii) For the quadrilateral, show that $9a + 7b = 310$.

[1]

- (iii) Solve these simultaneous equations.
Show all your working.

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots [3]$$

- (iv) Find the size of the smallest angle in the triangle.

$\dots\dots\dots [1]$

(b) Solve the equation $6x - 3 = -12$.

$x =$ [2]

(c) Rearrange $2(4x - y) = 5x - 3$ to make y the subject.

$y =$ [3]

(d) Simplify.

$$(27x^9)^{\frac{2}{3}}$$

..... [2]

(e) Simplify.

$$\frac{x^2 + 5x}{x^2 - 25}$$

..... [3]

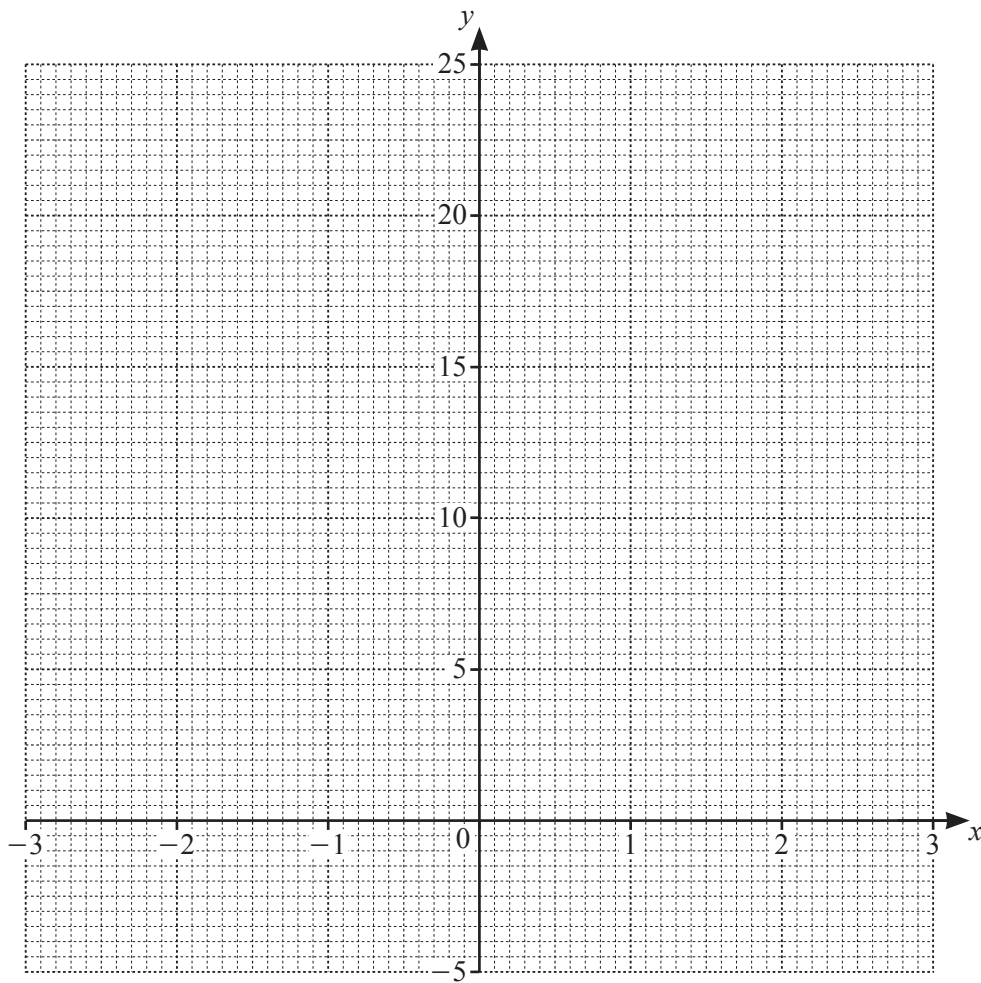
- 3 The table shows some values for $y = x^3 + x^2 - 5x$.

x	-3	-2	-1.5	-1	0	1	1.5	2	2.5	3
y	-3	6	6.4		0		-1.9	2	9.4	

(a) Complete the table.

[3]

(b) On the grid, draw the graph of $y = x^3 + x^2 - 5x$ for $-3 \leq x \leq 3$.



[4]

- (c) Use your graph to solve the equation $x^3 + x^2 - 5x = 0$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [2]

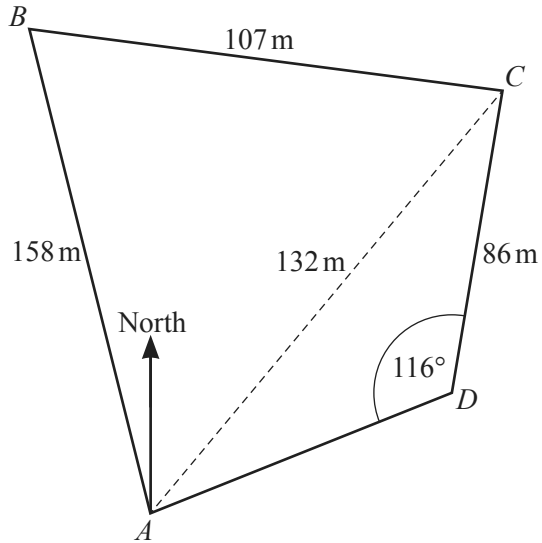
- (d) By drawing a suitable tangent, find an estimate of the gradient of the curve at $x = 2$.

$\dots\dots\dots$ [3]

- (e) Write down the largest value of the integer, k , so that the equation $x^3 + x^2 - 5x = k$ has three solutions for $-3 \leq x \leq 3$.

$k = \dots\dots\dots$ [1]

4

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The diagram shows a field, $ABCD$, on horizontal ground.

- (a) There is a vertical post at C .
From B , the angle of elevation of the top of the post is 19° .

Find the height of the post.

..... m [2]

- (b) Use the cosine rule to find angle BAC .

Angle $BAC =$ [4]

- (c) Use the sine rule to find angle CAD .

Angle $CAD = \dots\dots\dots$ [3]

- (d) Calculate the area of the field.

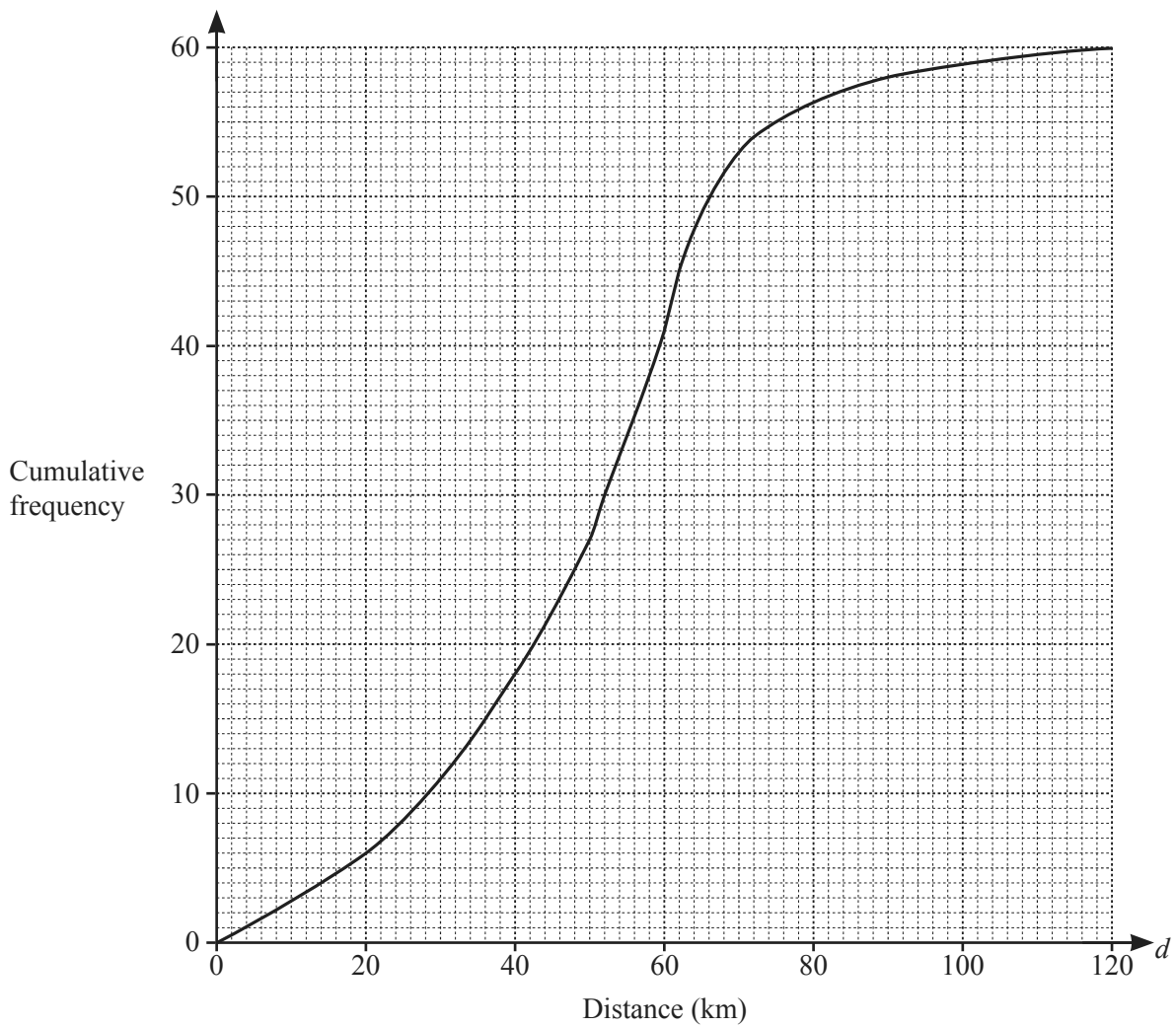
$\dots\dots\dots \text{m}^2$ [3]

- (e) The bearing of D from A is 070° .

Find the bearing of A from C .

$\dots\dots\dots$ [2]

- 5 The cumulative frequency diagram shows information about the distance, d km, travelled by each of 60 male cyclists in one weekend.



- (a) Use the cumulative frequency diagram to find an estimate of

(i) the median,

..... km [1]

(ii) the lower quartile,

..... km [1]

(iii) the interquartile range.

..... km [1]

- (b) For the same weekend, the interquartile range for the distances travelled by a group of female cyclists is 40 km.

Make one comment comparing the distribution of the distances travelled by the males with the distribution of the distances travelled by the females.

.....
 [1]

- (c) A male cyclist is chosen at random.

Find the probability that he travelled more than 50 km.

..... [2]

- (d) (i) Use the cumulative frequency diagram to complete this frequency table.

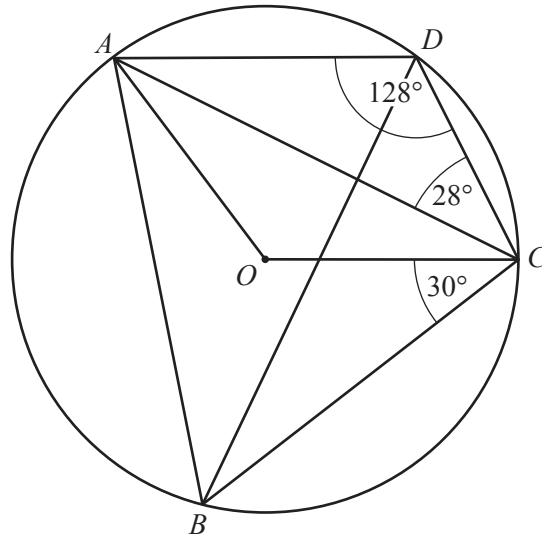
Distance (d km)	Number of male cyclists
$0 < d \leq 40$	18
$40 < d \leq 50$	9
$50 < d \leq 60$	
$60 < d \leq 70$	
$70 < d \leq 90$	
$90 < d \leq 120$	2

[2]

- (ii) Calculate an estimate of the mean distance travelled.

..... km [4]

6 (a)

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In the diagram, A , B , C and D lie on the circle, centre O .
 Angle $ADC = 128^\circ$, angle $ACD = 28^\circ$ and angle $BCO = 30^\circ$.

- (i) Show that obtuse angle $AOC = 104^\circ$.
 Give a reason for each step of your working.

[3]

- (ii) Find angle BAO .

Angle $BAO = \dots\dots\dots$ [2]

- (iii) Find angle ABD .

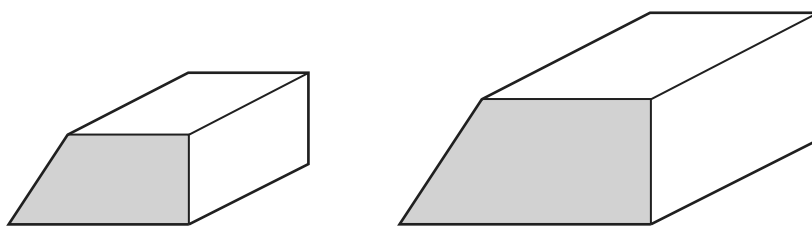
Angle $ABD = \dots\dots\dots$ [1]

- (iv) The radius, OC , of the circle is 9.6 cm.

Calculate the total perimeter of the sector $OADC$.

..... cm [3]

(b)



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The diagram shows two mathematically similar solid metal prisms.

The volume of the smaller prism is 648 cm^3 and the volume of the larger prism is 2187 cm^3 .

The area of the cross-section of the smaller prism is 36 cm^2 .

- (i) Calculate the area of the cross-section of the larger prism.

..... cm^2 [3]

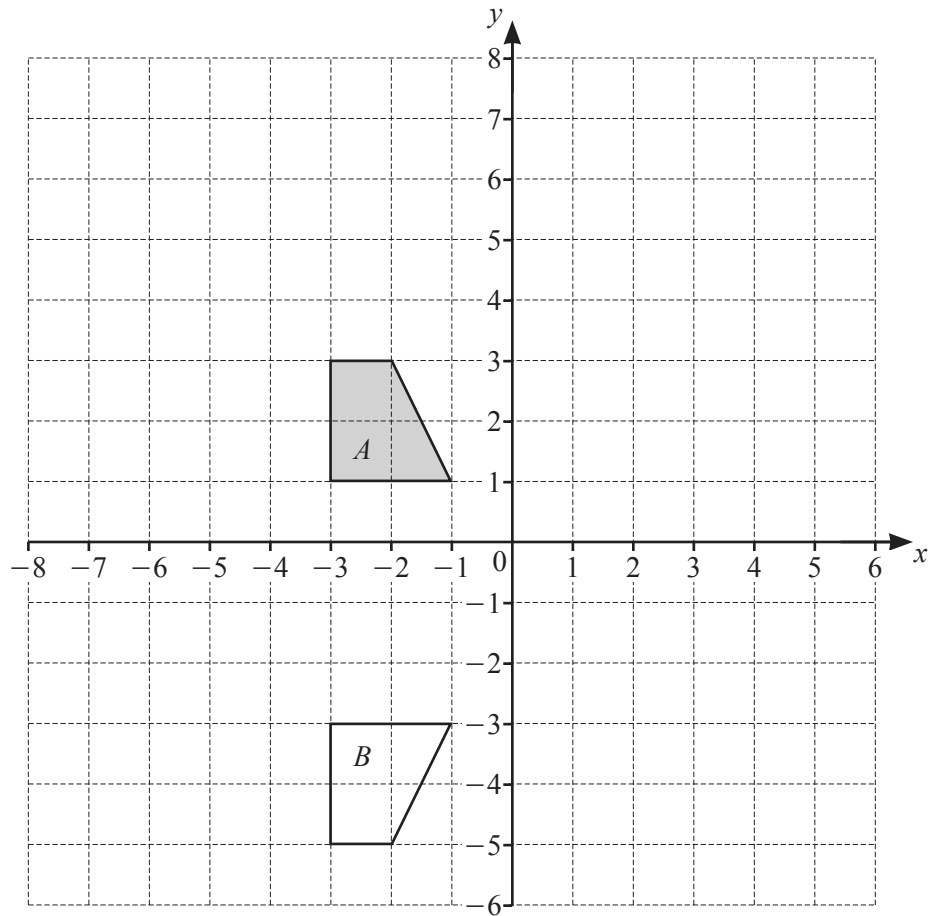
- (ii) The larger prism is melted down into a sphere.

Calculate the radius of the sphere.

[The volume, V , of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.]

..... cm [3]

7



- (a) Describe fully the **single** transformation that maps shape *A* onto shape *B*.

..... [2]

- (b) On the grid, draw the image of

- (i) shape *A* after a translation by the vector $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$, [2]
- (ii) shape *A* after a rotation through 180° about $(0, 0)$, [2]
- (iii) shape *A* after an enlargement, scale factor 2, centre $(-7, 0)$. [2]

- 8 (a) A bag contains 4 red marbles and 2 yellow marbles.
Behnaz picks two marbles at random without replacement.

Find the probability that

- (i) the marbles are both red,

..... [2]

- (ii) the marbles are not both red.

..... [1]

- (b) Another bag contains 5 blue marbles and 2 green marbles.
Bryn picks one marble at random without replacement.
If this marble is not green, he picks another marble at random without replacement.
He continues until he picks a green marble.

Find the probability that he picks a green marble on his first, second or third attempt.

..... [4]

9

$$f(x) = 2x - 3$$

$$g(x) = 9 - x^2$$

$$h(x) = 3^x$$

(a) Find

(i) $f(4)$,

..... [1]

(ii) $hg(3)$,

..... [2]

(iii) $g(2x)$ in its simplest form,

..... [1]

(iv) $fg(x)$ in its simplest form.

..... [2]

(b) Find $f^{-1}(x)$.

$$f^{-1}(x) = \dots\dots\dots [2]$$

(c) Find x when $5f(x) = 3$.

$$x = \dots\dots\dots [2]$$

(d) Solve the equation $gf(x) = -16$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

(e) Find x when $h^{-1}(x) = -2$.

$x = \dots\dots\dots$ [1]

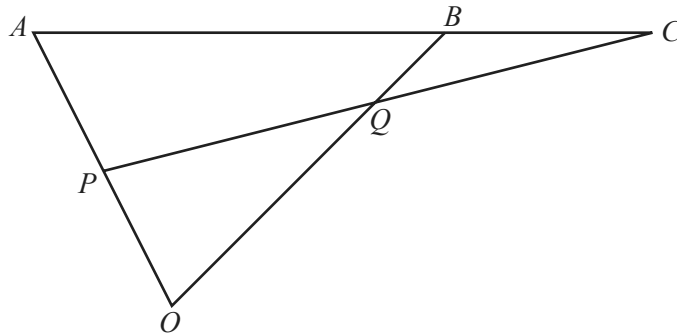
10 Solve.

$$\frac{1}{x} - \frac{2}{x+1} = 3$$

Show all your working and give your answers correct to 2 decimal places.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [7]

11

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OAB is a triangle and ABC and PQC are straight lines.

P is the midpoint of OA , Q is the midpoint of PC and $OQ : QB = 3 : 1$.
 $\vec{OA} = 4\mathbf{a}$ and $\vec{OB} = 8\mathbf{b}$.

(a) Find, in terms of $\mathbf{a}$ and/or $\mathbf{b}$, in its simplest form

(i) $\vec{AB}$,

$\vec{AB} = \dots\dots\dots$ [1]

(ii) $\vec{OQ}$,

$\vec{OQ} = \dots\dots\dots$ [1]

(iii) $\vec{PQ}$.

$\vec{PQ} = \dots\dots\dots$ [1]

(b) By using vectors, find the ratio $AB : BC$.

$\dots\dots\dots : \dots\dots\dots$ [3]

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